

Project Portfolio

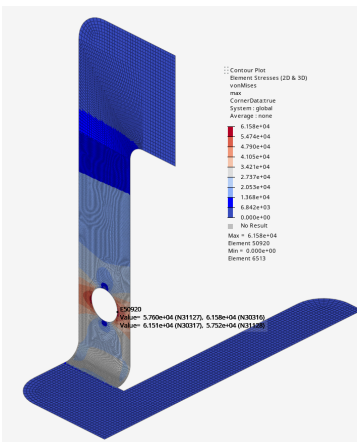


*Testing controller on Quanser
Qube-Servo 3*

Controls

Performed design of PID controllers in time and frequency domain. Designed full-state feedback and compensated full-state feedback controllers. Pictured full-state controller was designed to stabilize an inverted pendulum system with a full-state feedback controller built on Simulink.

[Details/Documentation](#)



*Von Mises stress contour on a
bookstop*

Finite Element Analysis

Analyzed a variety of shapes in HyperMesh and OptiStruct. Leveraging 2D and 3D elements, convergence studies, and buckling analysis, to make design recommendations on a variety of systems.

[Details/Documentation](#)

- The robotic attachment would essentially be a hand drill, that uses a "plug" of metal in the drill chuck rather than any drill bit. As for current SOA, NASA typically doesn't weld when they can avoid it, but if so they use TIG (Tungsten Inert Gas) Welding.
- Estimates are based on common Earth TIG Welding and hand drills.

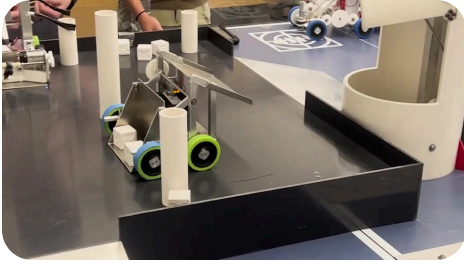
[illegible]

Principal Investigator

Linkage Development

Leveraged MSC Adams for multibody dynamic simulation to model the linkage system to hit all four targets. Designed transmission, links, and hardstops for a robust solution. Performed PID tuning to ensure the speed of performance and fast settling.

Details/Documentation

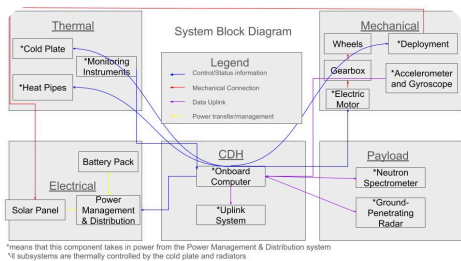


Robotic Machine Player confirming points for our team by delivering cubes into the delivery location

RMP Development

Developed CAD and design skills to complete multiple different tasks in a high-pressure competition. Designed robust transmission and powertrain systems. Executed analytical statics analysis to ensure the functionality of multiple functions simultaneously.

Details/Documentation



Full rover system block diagram

Lead Systems Engineer

Led a group of engineering students to a PDR for a lunar rover mission. Coordinated with the Science and Programmatic team on developing system-level requirements, and flowed these requirements down to Mechanical, Electrical, and Thermal requirements. Managed all interfacing between systems as well as protocol development.